NEXUS Alpha: the project in plain words

Overview for the owner, 7 November 2026. Code and full details: github.com/harshatangirala/nexus-alpha

What it is

NEXUS Alpha is a research system that looks at a US stock the way a trader does: three charts (weekly, daily, hourly), technical indicators, news, company filings, the company's financial statements and the wider market. It then estimates the chances that the stock goes up, sideways or down over a chosen period, from 1 day to 1 year, and turns that into a careful, rules-based trade plan. It is built to be honest: every result is checked against data the model never saw, and nothing is ever sent to a broker.

How it works

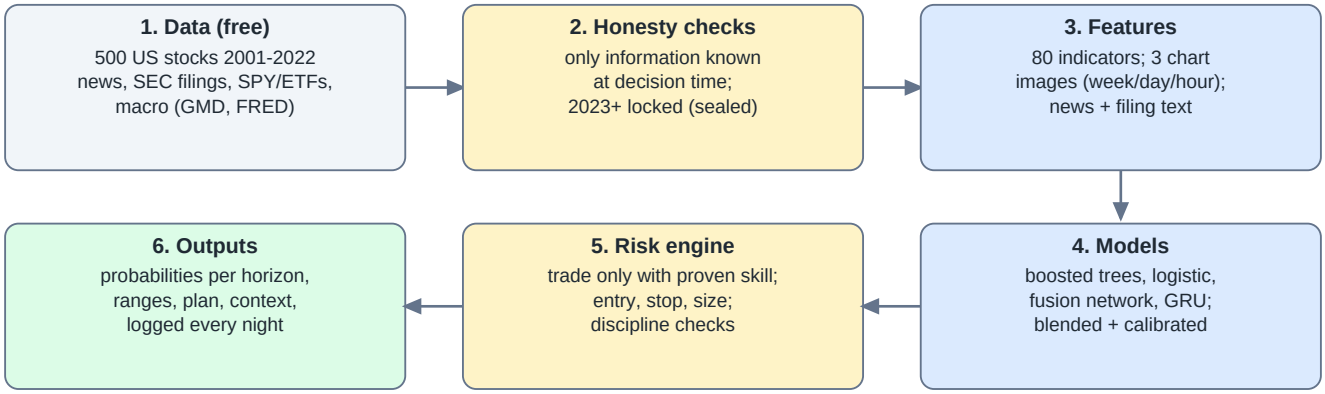

Figure 1. The whole system, from raw data to what you see.

Data comes only from free sources. Strict checks make sure the model only uses information that existed at the moment of the decision (09:00 New York, before the market opens). The years 2023 onward are locked away for one final exam. Features are built from prices, chart images, text and macro data; several model types are blended, calibrated, and passed to a risk engine that refuses to trade unless skill has been proven.

What you give and what you get

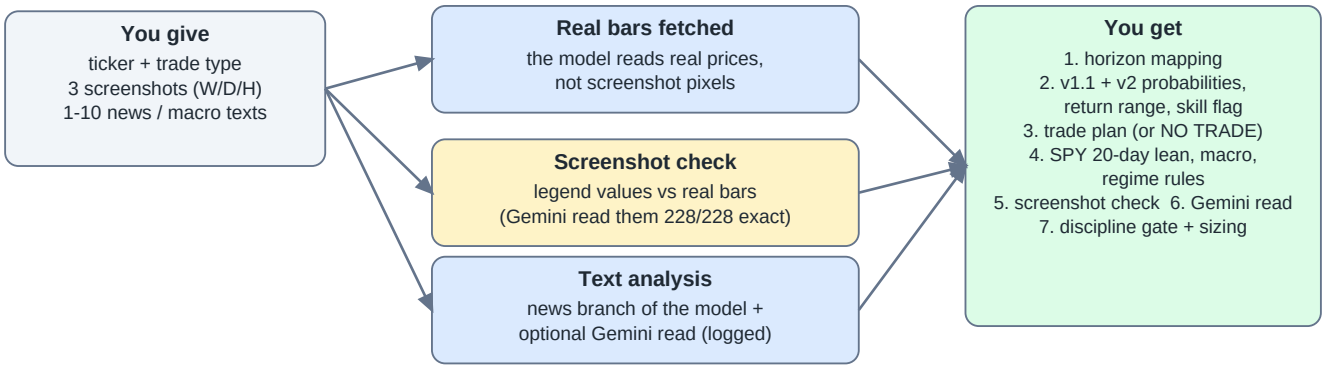

Figure 2. What happens on the Trade Desk when you upload screenshots and news.

Your screenshots are used to confirm the ticker and to check that the prices you see match the real data. The model itself reads clean, real price bars, because that is what it was trained on. Texts feed the news branch, and Gemini (optional, free tier) adds a written read that is logged so it can be scored later.

What each part contributes (measured, not assumed)

Part	What the tests found
Technical indicators	The strongest single input
Chart images (computer vision)	Small extra value
News and filings (NLP)	Weak; news exists for 46% of samples
Macro and S&P 500 data	No gain per stock; a useful SPY 20-day lean (AUROC 0.72 / 0.68)
Fundamentals (31 ratios, SEC)	Help ranking within each industry (E23 kept); in the forward test
Up / sideways / down	Better than the base rate at 1-20 days; direction itself is near a coin flip
Ranking stocks against each other	Real but small skill (rank IC 0.02-0.05); not yet shown to beat costs

In short: the system is good at judging how much a stock is likely to move, a little good at telling which stocks will do better than others, and not yet able to say which way a single stock will go. That is normal for markets, and it is why the project measures everything instead of trusting backtests.

Fundamental analysis

Every company's financial statements come from the SEC's free XBRL data (from 2009), exactly as they were known on each day: a restated number only counts once it was filed. From them the system computes 31 ratios (valuation such as earnings and free-cash-flow yield, margins, returns on capital, growth, debt and liquidity, efficiency, and quality scores: Piotroski, Altman Z and the earnings surprise), plus P/E, EV/EBITDA, P/S, P/B, P/FCF and a discounted cash flow you can adjust. The Trade Desk shows the three statements, every ratio with the stock's rank in the whole market and within its industry, and the DCF.

A strict data audit came first. It caught market values that were wrong by a split factor, by the ADR ratio of foreign companies, by share class, and by filings typed in thousands. All are now guarded, and the ten largest companies match their public values. Two pre-registered tests followed. Ratios ranked across the whole market did not pass the rule. Ranked within each industry, they did: they improved the ranking of stocks, mostly over months. That model (v3_fund) is frozen and joins the nightly forward test. It becomes a shown signal only if it beats the current ranker there.

How we keep it honest

Choices are made only on validation. Test is a reused check. Sealed and forward data are the real exams.

Figure 3. How the model is kept honest over time.

Every evening the frozen v1.1, v2 and v3_fund models forecast about 500 stocks, and the frozen SPY market read forecasts the market. The forecasts are hash-chained and pushed to the public repository before the market opens, so nobody can change them after the fact. A one-time sealed test of 2023 onward is ready; only you can open it.

The audit (SWOT, fishbone, Six Sigma)

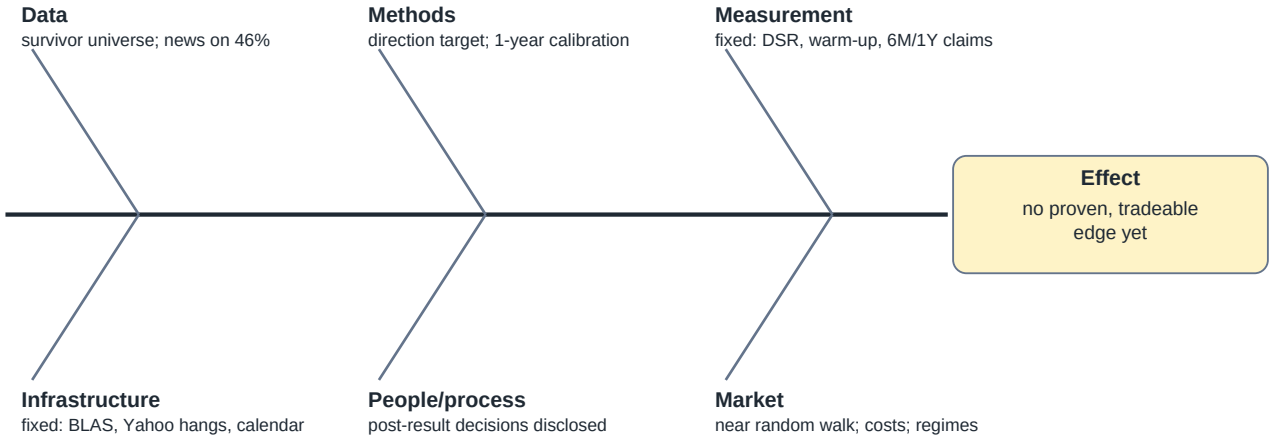

Figure 4. Fishbone (cause-and-effect) diagram from the audit.

Two audits were run back to back, following Six Sigma's Define-Measure-Analyze-Improve-Control steps. Six problems were found and fixed:

- The forward scorer's labels now match training 99.2% of the time (was 98.4%).
- A broken return column in the v2 data was repaired.
- The decision-time rule for live forecasts beyond the calendar was corrected.
- The dashboard no longer claims skill at 6 months and 1 year.
- The calendar was extended to 2030.
- The backtest is now reported with its deflated Sharpe ratio (about 0, so no proven profit).

A new test (E21) tried three dedicated stock rankers; none beat the existing model, which now has its rankings scored every night. A third and fourth audit (fundamentals and the whole project) found 15 more issues and fixed 12 (the other 3 are documented data limits), including a Yahoo outage guard for the nightly run and a test that checks the nightly script itself. Control charts and the full test suite (109 tests, all passing) run every night.

Is it complete without paid subscriptions?

Yes. Everything runs on free data and a normal CPU. Gemini, QVeris and a GPU are optional extras.

What is left

- **You:** sign in to QVeris (/mcp) if you want its data.
- **You:** open the sealed 2023+ test once, when ready (commands in docs/SEALED_EVALUATION_PLAN.md, 7b; about 3-5 hours on CPU).
- **Optional:** add two retry times to the nightly task (one command in docs/AUDIT_3_4_2026-11-07.md, section 5).
- **Time:** let the nightly forward test run. About 20 sessions before the control charts mean anything, and about 6 months (120 sessions) before v3_fund's fundamentals ranking can be confirmed or rejected.
- **Next research (pre-registered):** a live news feed variant, a longer calibration window, an earnings-drift test, and the stock setups from the trading-skill libraries.
- **Rule:** no real trading decisions until the forward test and the sealed test both show skill after costs.